

HOUSTON DUNN, TX, USA

WMO: 720594

Lat: 29.717N

Lon: 95.383W

Elev: 227

StdP: 14.58

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00188

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.0	36.3	13.7	11.1	42.2	17.8	13.6	47.1	24.6	51.1	22.6	53.7	10.5	350	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.3	99.8	77.0	98.5	76.5	95.2	76.2	79.5	90.2	79.1	89.4	78.5	88.0	4.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	143.9	81.9	77.2	142.7	81.8	76.8	140.7	81.8	43.2	89.0	42.6	89.7	42.1	87.4	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.5	19.0	17.2	28.2	103.6	4.8	4.4	24.8	106.7	22.0	109.3	19.3	111.7	15.8	114.9
			23.9	80.9	4.3	0.9	20.8	81.6	18.3	82.1	15.8	82.7	12.7	83.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	72.2	55.1	59.5	65.2	71.2	77.9	84.4	86.2	87.5	82.4	74.5	64.1	57.6
	DBStd	13.24	9.67	10.23	8.31	5.92	4.92	3.47	3.03	3.17	4.32	7.27	9.03	9.63
	HDD50	153	61	36	8	0	0	0	0	0	0	11	37	
	HDD65	1055	325	200	101	20	2	0	0	0	16	122	269	
	CDD50	8258	220	302	479	636	864	1031	1122	1163	973	760	435	273
	CDD65	3682	19	46	106	206	400	581	657	698	523	311	96	39
	CDH74	36860	53	156	468	1174	3345	6677	8003	8671	5282	2459	446	126
	CDH80	16348	1	19	91	260	1164	3178	3957	4517	2221	847	83	10
Wind	WSAvg	7.7	8.5	9.4	9.3	9.6	8.2	6.9	5.5	5.4	6.1	6.9	8.2	8.5
Precipitation	PrecAvg	42.7	3.2	2.6	2.5	2.9	4.4	5.0	3.6	4.0	4.8	3.3	3.2	3.0
	PrecMax	62.2	6.7	5.7	6.1	10.0	11.0	15.0	8.3	10.6	14.9	11.1	7.2	7.1
	PrecMin	22.9	0.3	0.4	0.6	0.3	0.7	0.4	0.7	0.9	0.8	0.0	0.4	0.4
	PrecStd	10.0	1.8	1.4	1.4	2.5	2.5	4.3	1.7	2.6	2.9	2.8	2.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.7	82.2	86.1	89.7	93.5	100.2	100.2	105.7	100.0	93.0	86.2	81.5
		MCWB	65.3	67.7	68.9	66.9	72.3	76.4	77.0	77.8	75.9	75.0	73.1	69.0
	2%	DB	75.1	78.9	82.1	85.6	90.7	97.2	98.9	100.4	96.5	90.5	81.9	77.1
		MCWB	64.2	67.5	68.0	68.2	73.4	75.7	76.6	77.8	74.9	75.1	70.3	68.7
	5%	DB	72.3	75.0	79.2	82.3	88.3	94.8	96.5	99.0	93.1	87.9	79.0	74.7
		MCWB	63.7	66.1	66.5	68.2	73.2	75.8	76.3	77.3	75.1	74.0	68.9	68.5
	10%	DB	68.4	72.4	75.5	81.1	86.0	92.6	94.6	95.2	90.7	84.4	75.4	72.3
		MCWB	62.7	65.7	65.2	67.8	72.4	75.8	76.5	76.8	74.8	71.1	68.0	67.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.7	72.2	72.5	75.4	78.0	80.2	80.0	80.5	79.4	79.2	75.7	73.6
		MCDB	72.3	75.3	78.1	81.1	85.6	88.6	89.6	91.4	89.5	84.8	81.0	76.0
	2%	WB	68.6	70.5	71.4	74.1	76.9	79.1	79.2	79.8	78.5	78.0	73.9	72.1
		MCDB	71.7	74.3	77.4	79.1	84.5	88.0	89.0	91.6	87.5	84.2	78.2	75.1
	5%	WB	66.9	69.4	70.3	73.0	76.1	78.4	78.5	79.2	78.1	76.7	72.5	70.3
		MCDB	70.3	73.2	75.3	77.8	83.5	87.1	87.8	91.0	86.5	83.2	76.9	73.8
	10%	WB	64.2	68.0	69.3	71.9	75.2	77.9	78.2	78.6	77.4	75.5	70.4	68.3
		MCDB	66.7	71.3	73.6	77.1	82.2	86.4	87.3	89.2	85.2	81.9	75.1	71.0
Mean Daily Temperature Range	5% DB	MDBR	16.3	15.7	16.1	16.1	15.6	16.8	16.9	18.3	15.8	17.2	16.2	15.2
		MCDBR	18.5	16.7	17.8	17.8	17.7	19.0	19.0	21.7	19.0	17.6	16.4	16.8
		MCWBR	11.4	9.5	8.3	7.4	5.2	4.6	4.0	4.5	5.1	6.0	8.5	11.2
	5% WB	MCDBR	16.1	15.3	15.2	14.3	14.6	16.4	16.8	19.0	15.8	15.6	15.6	16.6
	MCWBR	13.4	11.4	9.2	8.2	5.1	4.6	4.2	4.5	4.7	8.0	9.5	13.5	
Clear-Sky Solar Irradiance	taub	0.339	0.357	0.371	0.435	0.478	0.468	0.495	0.483	0.453	0.378	0.369	0.347	
	taud	2.505	2.464	2.416	2.237	2.153	2.234	2.179	2.214	2.263	2.464	2.438	2.489	
	Ebn at Noon	284	288	290	274	261	262	255	257	261	277	270	275	
	Edh at Noon	29	32	36	45	49	45	47	45	42	32	30	28	
All-Sky Solar Radiation	RadAvg	912	1027	1406	1681	1869	1942	1899	1822	1577	1372	1034	809	
	RadStd	88	179	121	120	136	157	178	149	165	166	105	97	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+200	+175

Nomenclature: See separate page